

How We Processed the Bonn Dataset

Its 5 Subsets and How They Became a Binary Task

EEGdiff_V2 project

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1 The Bonn dataset has 5 subsets

The classic Bonn University EEG database (Andrzejak et al., 2001) ships as **5 sets**, labeled **Z, O, N, F, S** — 100 single-channel recordings each (23.6 s / 4097 samples @ 173.61 Hz), 500 recordings total. Each set is a different clinical condition:

Set (file prefix)	What it actually is
Z	Healthy volunteers, eyes open , scalp EEG
O	Healthy volunteers, eyes closed , scalp EEG
N	Epilepsy patients, seizure-free interval , intracranial electrodes in the hippocampus opposite to the epileptic focus
F	Epilepsy patients, seizure-free interval , intracranial electrodes at the epileptic focus
S	Epilepsy patients, recorded during an actual seizure (ictal activity)

We verified this directly against the raw data: each of the 5 subset zip files (**z.zip**, **o.zip**, **n.zip**, **f.zip**, **s.zip**) extracts to exactly 100 real **.txt** files (macOS resource-fork junk excluded), 500 files total.

2 How we mapped 5 subsets onto our binary task

Our task is binary seizure-vs-normal, so `prepare_bonn()` in `dataset_tools/prepare_external_eeg.py` collapses the 5 subsets into 2 classes with one simple rule:

```
label = "seizure" if stem.startswith("S") or parent.startswith("S") else "normal"
```

In plain terms

Only **Set S** is labeled **"seizure"**. Sets **Z, O, N, and F** are all lumped together as **"normal"** — regardless of whether they are healthy controls (Z, O) or epilepsy patients who simply are not seizing right now (N, F).

We verified the final counts directly from the processed metadata: each of the 5 subsets contributed exactly 100 windows (matching the 100 raw files per set), and the resulting train/test split is:

Split	normal	seizure
train	320	80
test	80	20
total	400	100

That $400/100 = 4:1$ ratio is exactly “4 non-seizure subsets ($Z+O+N+F=400$) vs. 1 seizure subset ($S=100$).”

3 Worth remembering

- Our "normal" class is actually a mix of three quite different things: healthy eyes-open, healthy eyes-closed, and epilepsy patients who are currently seizure-free (sets N and F) — all treated identically as *not seizure*. Clinically, N and F are the **hardest negatives** (epileptic brain tissue, just not actively seizing), so lumping them with healthy controls is a real simplification of the task.
- This is also why Bonn is a comparatively **easy** benchmark in the literature: the actual discriminative signal (ictal S vs. everything else) is quite large. Consistent with this, our own result on Bonn was very strong (ROC-AUC 0.993).
- Each Bonn recording is a **single channel** with no electrode name at all (a separate fact from which subset it belongs to). It is placed into cup #1 (FP1) of our 22-channel frame, with the other 21 channels zero-filled — see the companion handout *How We Mapped Every EEG Dataset to 22 Channels*.